Applied Database Technologies

Lab 6: Functions

Chapter : How to code subqueries

1. Write a SELECT statement that returns these columns from the Products table:

   The list_price column

   The discount_percent column

   A column named discount_amount that uses the previous two columns to calculate the discount amount and uses the ROUND function to round the result so it has 2 decimal digits

list_price	discount_percent	discount_amount
699.00	30.00	209.70
1199.00	30.00	359.70
2517.00	52.00	1308.84
489.99	38.00	186.20
299.00	0.00	0.00
415.00	39.00	161.85
799.99	30.00	240.00
499.99	25.00	125.00
699.99	30.00	210.00
799.99	15.00	120.00

   SELECT list_price, discount_percent,

   ROUND((list_price *(discount_percent / 100)), 2) AS discount_amount

   FROM products p;

2. Write a SELECT statement that returns these columns from the Orders table:

   The order_date column

   A column that uses the DATE_FORMAT function to return the four-digit year that's stored in the order_date column

A column that uses the DATE_FORMAT function to return the order_date column in this format: Mon-DD-YYYY. In other words, use abbreviated months and separate each date component with dashes.

A column that uses the DATE_FORMAT function to return the order_date column with only the hours and minutes on a 12-hour clock with an am/pm indicator

A column that uses the DATE_FORMAT function to return the order_date column in this format: MM/DD/YY HH:MM. In other words, use two-digit months, days, and years and separate them by slashes. Use 2-digit hours and minutes on a 24-hour clock. And use leading zeros for all date/time components.

order_date	format_1	format_2	format_3	format_4
2018-03-28 09:40:28	2018	Mar-28-2018	09:40 AM	03/28/18 09:40
2018-03-28 11:23:20	2018	Mar-28-2018	11:23 AM	03/28/18 11:23
2018-03-29 09:44:58	2018	Mar-29-2018	09:44 AM	03/29/18 09:44
2018-03-30 15:22:31	2018	Mar-30-2018	03:22 PM	03/30/18 15:22
2018-03-31 05:43:11	2018	Mar-31-2018	05:43 AM	03/31/18 05:43
2018-03-31 18:37:22	2018	Mar-31-2018	06:37 PM	03/31/18 18:37
2018-04-01 23:11:12	2018	Apr-01-2018	11:11 PM	04/01/18 23:11
2018-04-02 11:26:38	2018	Apr-02-2018	11:26 AM	04/02/18 11:26
2018-04-03 12:22:31	2018	Apr-03-2018	12:22 PM	04/03/18 12:22

```
SELECT order_date,
DATE_FORMAT(order_date, '%Y' ) as format_1,
DATE_FORMAT(order_date, '%b- %d-%Y') as format_2,
DATE_FORMAT(order_date, '%h:%i %p') as format_3,
DATE_FORMAT(order_date, '%m/%d/%Y %H:%i') as format_4
FROM orders o;
```

3. Write a SELECT statement that returns these columns from the Orders table:

   The order_id column

The order_date column

A column named approx_ship_date that's calculated by adding 2 days to the order_date column

The ship_date column if it doesn't contain a null value

A column named days_to_ship that shows the number of days between the order date and the ship date

When you have this working, add a WHERE clause that retrieves just the orders for March 2018.

Hint: Consider the DATEDIFF() and IFNULL() functions

order_id	order_date	approx_ship_date	IFNULL(ship_date, '')	days_to_ship
1	2018-03-28 09:40:28	2018-03-30 09:40:28	2018-03-30 15:32:51	2
2	2018-03-28 11:23:20	2018-03-30 11:23:20	2018-03-29 12:52:14	1
3	2018-03-29 09:44:58	2018-03-31 09:44:58	2018-03-31 09:11:41	2
4	2018-03-30 15:22:31	2018-04-01 15:22:31	2018-04-03 16:32:21	4
5	2018-03-31 05:43:11	2018-04-02 05:43:11	2018-04-02 14:21:12	2
6	2018-03-31 18:37:22	2018-04-02 18:37:22		1837

SELECT order_id, order_date,

DATE_ADD(order_date, INTERVAL 2 DAY) as approx_ship_date,

IFNULL(ship_date, ' '),

DATEDIFF(ship_date, order_date) as days_to_ship

FROM orders o

WHERE MONTH(order_date) = '03' AND YEAR(order_date) = '2018';

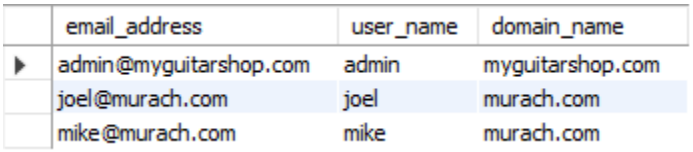
4. Write a SELECT statement that uses regular expression functions to get the username and domain name parts of the email addresses in the Administrators table. Return these columns:

   The email_address column

   A column named user_name that contains the username part of the email_address column (the part before the @ symbol)

   A column named domain_name that contains the domain name part of the email_address column (the part after the @ symbol

   Note: The username part of the email addresses contains only letters, and the domain name part contains only letters and a period. If regular expressions are too difficult, use substring_index() function rather than regular expressions.



email_address	user_name	domain_name
admin@myguitarshop.com	admin	myguitarshop.com
joel@murach.com	joel	murach.com
mike@murach.com	mike	murach.com

SELECT email_address,

SUBSTRING_INDEX(email_address, '@' , 1) as user_name,

SUBSTRING_INDEX(email_address, '@', -1) as domain_name

FROM administrators a;